PYTHON FOUNDATIONS - ASSIGNMENT 05

Mixed Operators & Membership Challenges

10 scenario-based Python challenges combining arithmetic, logical, comparison and membership operators in real-world decision making.

in | not in | and | or | not | + - * / % | if / elif / else

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK

10 MULTI-PART CHALLENGES

Quick reference before you begin

These questions intentionally mix earlier topics.

Do not solve them by memorizing syntax - first identify the arithmetic, condition and membership check required.

Arithmetic

+ - * / // % **

calculate values

Logical

and or not

combine/reverse conditions

Membership

in not in

check whether a value belongs

1 **Calculate first**

Example: total = price * quantity before checking affordability.

2 **Use membership for allowed/blocked values**

Example: city in allowed_cities.

3 **Combine conditions carefully**

Example: course in courses and fee_paid >= fee.

4 **Use parentheses for mixed logic**

Example: age >= 18 and (role in roles or has_pass).

Important: Use input, type conversion, arithmetic, comparisons, membership and if/elif/else. No functions required.

Membership + arithmetic + mandatory conditions

Read the scenario, identify the operators, then write the program.

1 Smart Course Registration

IN AND ARITHMETIC

A training institute accepts registrations only for courses currently offered and only when the payment requirements are satisfied.

1. Create available_courses = ["python", "java", "sql", "playwright"].
2. Take course name, course fee and amount already paid from the user.
3. Calculate the remaining fee using course_fee - amount_paid.
4. Registration is successful only if the course is in available_courses AND amount_paid is at least 50% of the course fee.
5. If the course is valid but payment is insufficient, print how much more is needed to reach 50%.
6. If the course is not available, print Course not available.

2 Online Shopping Coupon Challenge

IN OR * / IF-ELIF-ELSE

An online store provides discounts based on cart value, membership and valid coupon codes.

1. Create valid_coupons = ["SAVE10", "SAVE20", "NEWUSER"].
2. Take item price, quantity, coupon code and membership type from the user.
3. Calculate subtotal = price * quantity.
4. If subtotal >= 5000 AND coupon is in valid_coupons, apply 20% discount.
5. Else if subtotal >= 3000 OR membership is premium, apply 10% discount.
6. Otherwise apply no discount. Print subtotal, discount amount and final bill.

Calculate first, then validate with membership and logic

Read the scenario, identify the operators, then write the program.

3 Exam Eligibility & Subject Validation

IN AND AVERAGE

A student can sit for the final assessment only for approved subjects and only when both performance and attendance requirements are met.

1. Create subjects = ["maths", "science", "english", "computer"].
2. Take subject, three test marks and attendance percentage.
3. Calculate average marks.
4. Student is eligible only if subject is in subjects AND average >= 40 AND attendance >= 75.
5. If subject is valid but average is below 40, print how many more marks are needed to reach 40 average.
6. If attendance is below 75, also print the attendance shortage.

4 Travel Booking Budget Checker

IN AND -

A travel portal allows bookings only for supported destinations and when the traveller has enough money.

1. Create destinations = ["goa", "delhi", "jaipur", "mumbai"].
2. Take destination, number of travellers, ticket price per person and wallet balance.
3. Calculate total_cost = travellers * ticket_price.
4. Booking succeeds only if destination is in destinations AND wallet balance >= total_cost.
5. If destination is valid but balance is low, calculate and print the shortage.
6. If destination is not supported, print Destination unavailable.

Use not in, floor division and layered conditions

Read the scenario, identify the operators, then write the program.

5 Warehouse Packing & Restricted Items

NOT IN // % AND

A warehouse packs allowed products into boxes but must reject restricted products.

1. Create restricted_items = ["chemical", "battery", "aerosol"].
2. Take product name, total quantity and box capacity.
3. Continue only if product is not in restricted_items AND quantity > 0 AND box_capacity > 0.
4. Calculate complete boxes using // and leftover items using %.
5. Print complete boxes and leftovers.
6. If product is restricted, print Shipment blocked.

6 Employee Bonus & Department Check

IN AND / OR %

A company awards a special bonus only to employees in selected departments who also satisfy performance rules.

1. Create eligible_departments = ["qa", "engineering", "devops"].
2. Take department, monthly salary, performance score and years of experience.
3. Calculate annual salary = monthly_salary * 12.
4. Give a 10% bonus if department is in eligible_departments AND performance >= 80 AND experience >= 2.
5. Give a 5% bonus if department is eligible AND (performance >= 70 OR experience >= 5).
6. Print annual salary, bonus amount and final annual compensation.

Combine membership checks with age, budget and eligibility

Read the scenario, identify the operators, then write the program.

7 Secure Login & Role Access

IN / NOT IN | AND | NOT

An application allows login only for recognized roles and must reject blocked usernames.

1. Create allowed_roles = ["admin", "manager", "tester"] and blocked_users = ["guest", "demo", "temp"].
2. Take username, role and account balance from the user. Convert username and role to lowercase.
3. Access is allowed only if role is in allowed_roles AND username is not in blocked_users AND balance >= 0.
4. If username is blocked, print Blocked account.
5. If role is invalid, print Unauthorized role.
6. Otherwise print the reason the access check failed.

8 Cinema Booking & Age Rules

IN | AND / OR | *

A cinema sells tickets for available movie categories, but some categories have age restrictions.

1. Create categories = ["comedy", "drama", "action", "horror"].
2. Take category, age, ticket price, number of tickets and wallet balance.
3. Calculate total price.
4. Booking is possible only if category is in categories AND wallet balance >= total price.
5. For horror, age must also be >= 18. For all other listed categories, age must be >= 13.
6. Print either Booking confirmed or the exact reason it failed.

Bring arithmetic, logical and membership operators together

Read the scenario, identify the operators, then write the program.

9 Delivery Zone & Free Shipping

IN | AND / OR | -

A food delivery app checks whether an address is serviceable and whether the order qualifies for free delivery.

1. Create service_zones = ["madhapur", "gachibowli", "kondapur", "hitech city"].
2. Take delivery area, food total, delivery charge and premium-member status.
3. Order can proceed only if area is in service_zones AND food total > 0.
4. Free delivery applies if food total >= 1000 OR premium member is yes.
5. Otherwise add the delivery charge to the food total.
6. Print serviceability, delivery fee charged and final payable amount.

10 Automation Lab Access - Master Challenge

IN / NOT IN | AND / OR / NOT | ARITHMETIC

A training lab grants access based on course enrollment, account status, attendance and available lab credits.

1. Create enrolled_courses = ["python", "playwright", "pytest"] and blocked_users = ["guest", "suspended"].
2. Take username, selected course, attendance, lab credits and number of sessions requested.
3. Each session costs 2 credits. Calculate required_credits = sessions * 2 and remaining_credits after booking.
4. Access is allowed only if course is in enrolled_courses AND username is not in blocked_users AND attendance >= 75 AND lab_credits >= required_credits.
5. If credits are insufficient, print how many more credits are needed.
6. If access is allowed, print remaining credits. Otherwise print every applicable reason for rejection.